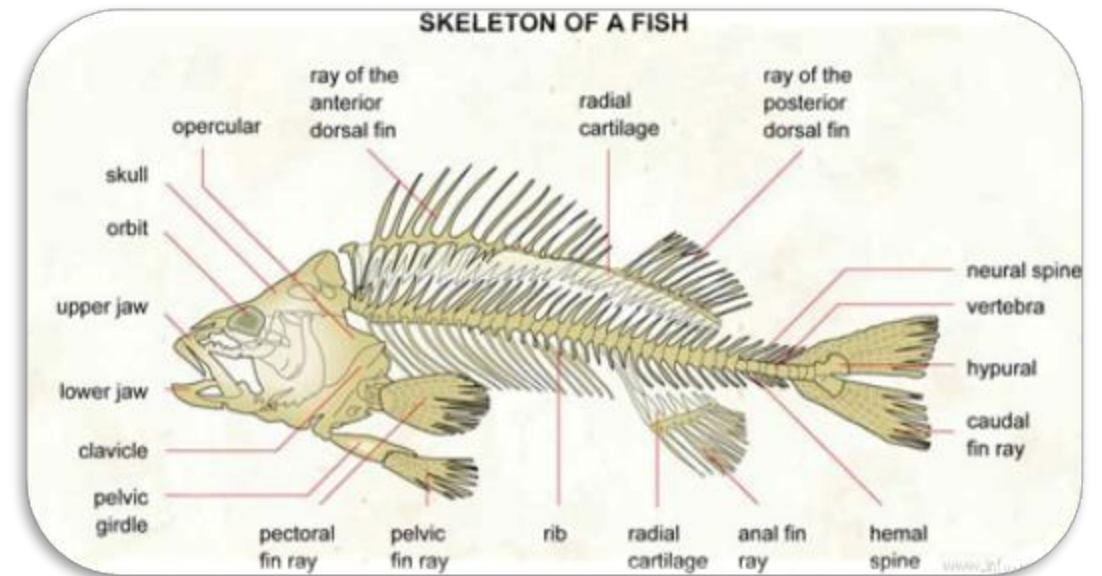
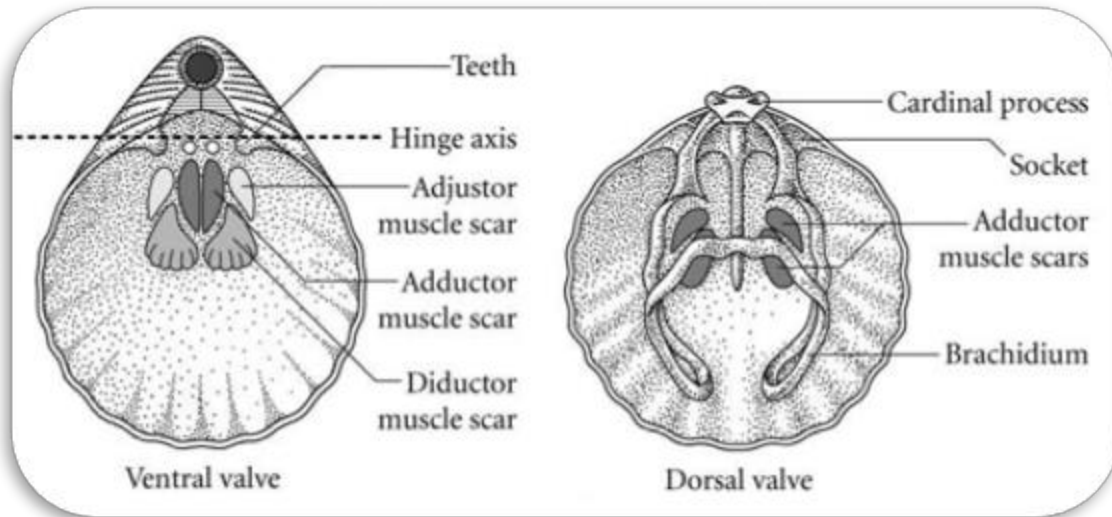


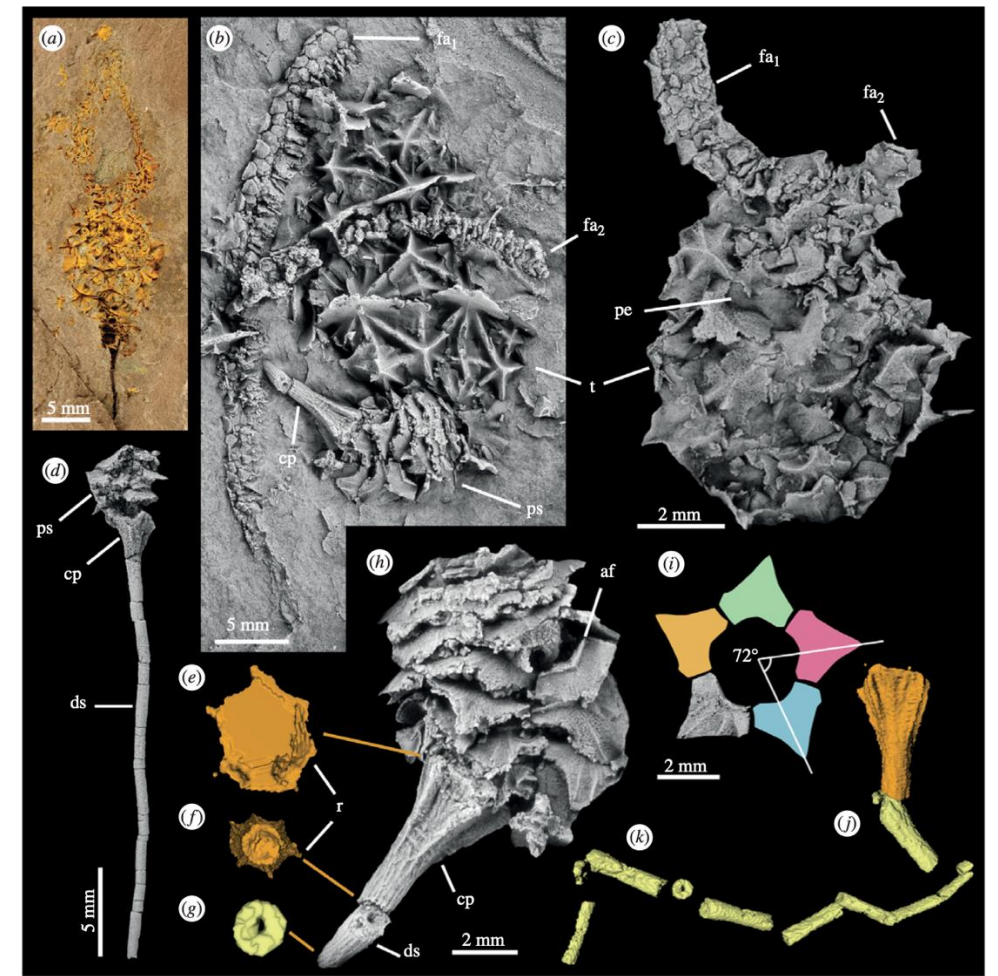
Challenges with morphological data

Morphological matrices are often quite small:

- Collection is very time consuming
- Number of characters available can be very small depending on the group



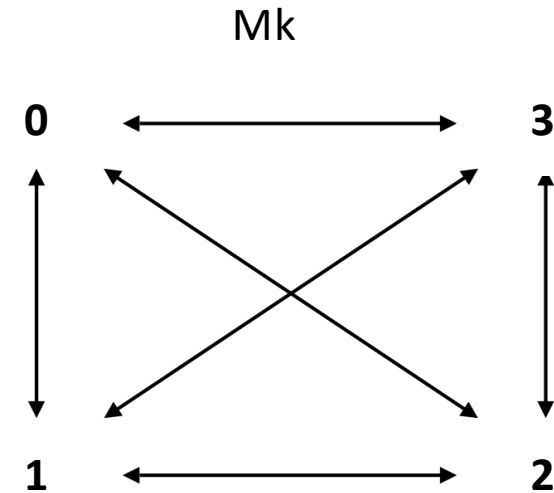
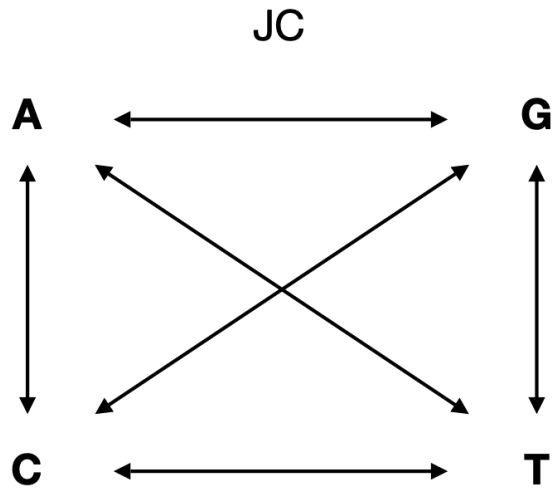
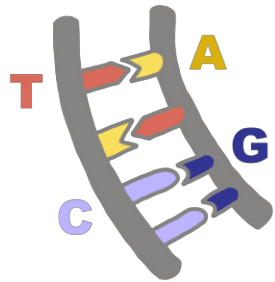
	Appendage branching pattern					Coring plate arrangement					Presence Absence				
001510010?00-100--00000000000															
000500010?200100--00100100000															
002500010?200100--0?100100000															
00?5?0010?200100?-0??010110															
0015000101201000430100011111															
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??050? ??? ?201000440?11011111															
01050?010-210000?501??010110															
00020001002101003-1110010110															
0002000100211001441121011111															
000201111-210010?-??11011121															
?103?0?11?1001104-0000010000															
1005002110100010--0?00110?20															
1005002000101010540?00110020															



Dibrachicystis purujoensis

Cambrian stalked echinoderms show unexpected plasticity of arm construction
Zamora & Smith. 2012. Proc B

Substitution models for morphological data

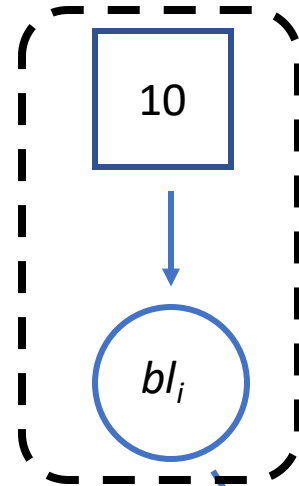


*4 state here as an example, can be any number from 2!

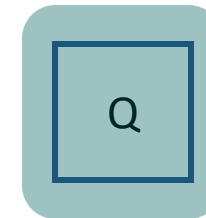
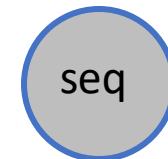
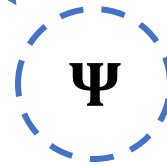
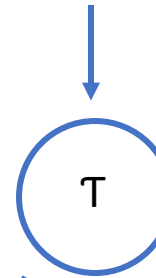
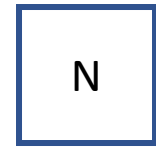
Line width represents the relative rate of change between different steps.

Mk Model

Exponential rate parameter of 10 on branch lengths.



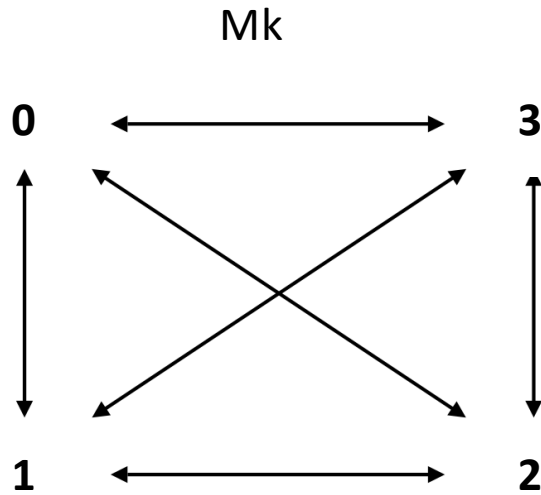
Uniform topology of N taxa



Mk model



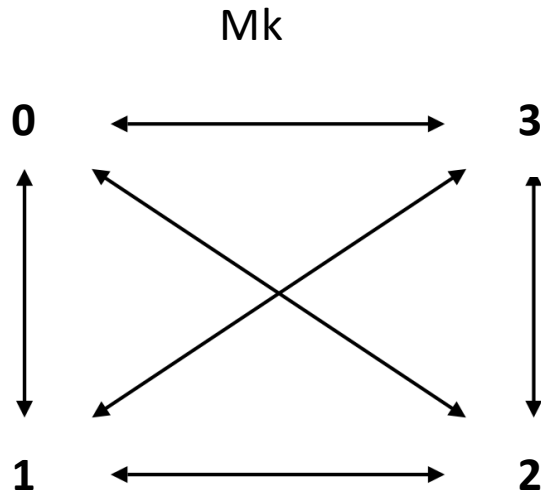
Substitution models for morphological data



$$Q = \begin{pmatrix} -\mu_0 & \mu_{01} & \mu_{02} & \mu_{03} \\ \mu_{10} & -\mu_1 & \mu_{12} & \mu_{13} \\ \mu_{20} & \mu_{21} & -\mu_2 & \mu_{23} \\ \mu_{30} & \mu_{31} & \mu_{32} & -\mu_3 \end{pmatrix},$$

*4 state here as an example, can be any number from 2!

Substitution models for morphological data

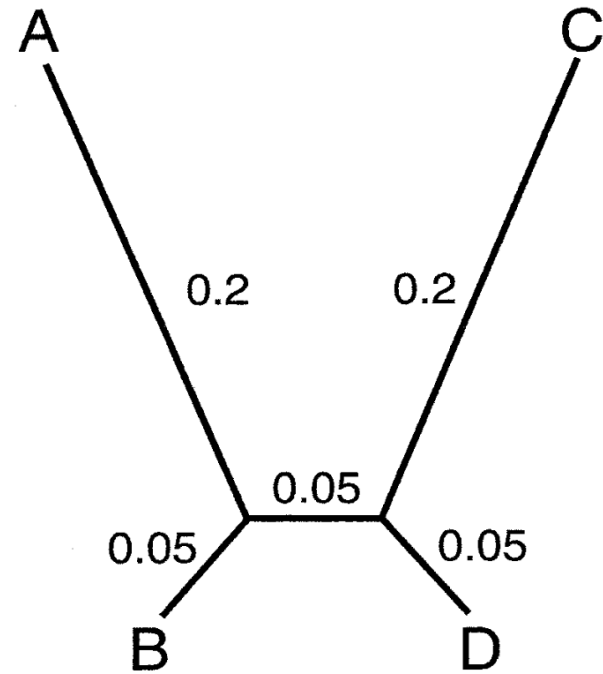


We can **add extensions** to the standard Mk model in a number of ways

*4 state here as an example, can be any number from 2!

Ascertainment Bias (V)

Conditions on the fact that all sites are variable



	True branch length	Mk (uncorrected)	Mkv (corrected)
Percent correct	—	74.0	99.8
Branch A	0.2	241,750 ($\pm 349,100$)	0.206 (± 0.060)
Branch B	0.05	0.43210 (± 0.13756)	0.050 (± 0.018)
Branch X	0.05	54.646 ($\pm 1,725.3$)	0.052 (± 0.023)
Branch C	0.2	143,950 ($\pm 228,910$)	0.206 (± 0.059)
Branch D	0.05	0.022 (± 0.054)	0.051 (± 0.019)

Across Site Rate Variation (+G)

	T1	T2
Taxa A	0	0
Taxa B	0	1
Taxa C	1	2

The transition rate
will impact branch
lengths

Slow rate of evolution



Fast rate of evolution

Relative to each other!

Across Site Rate Variation (+G)

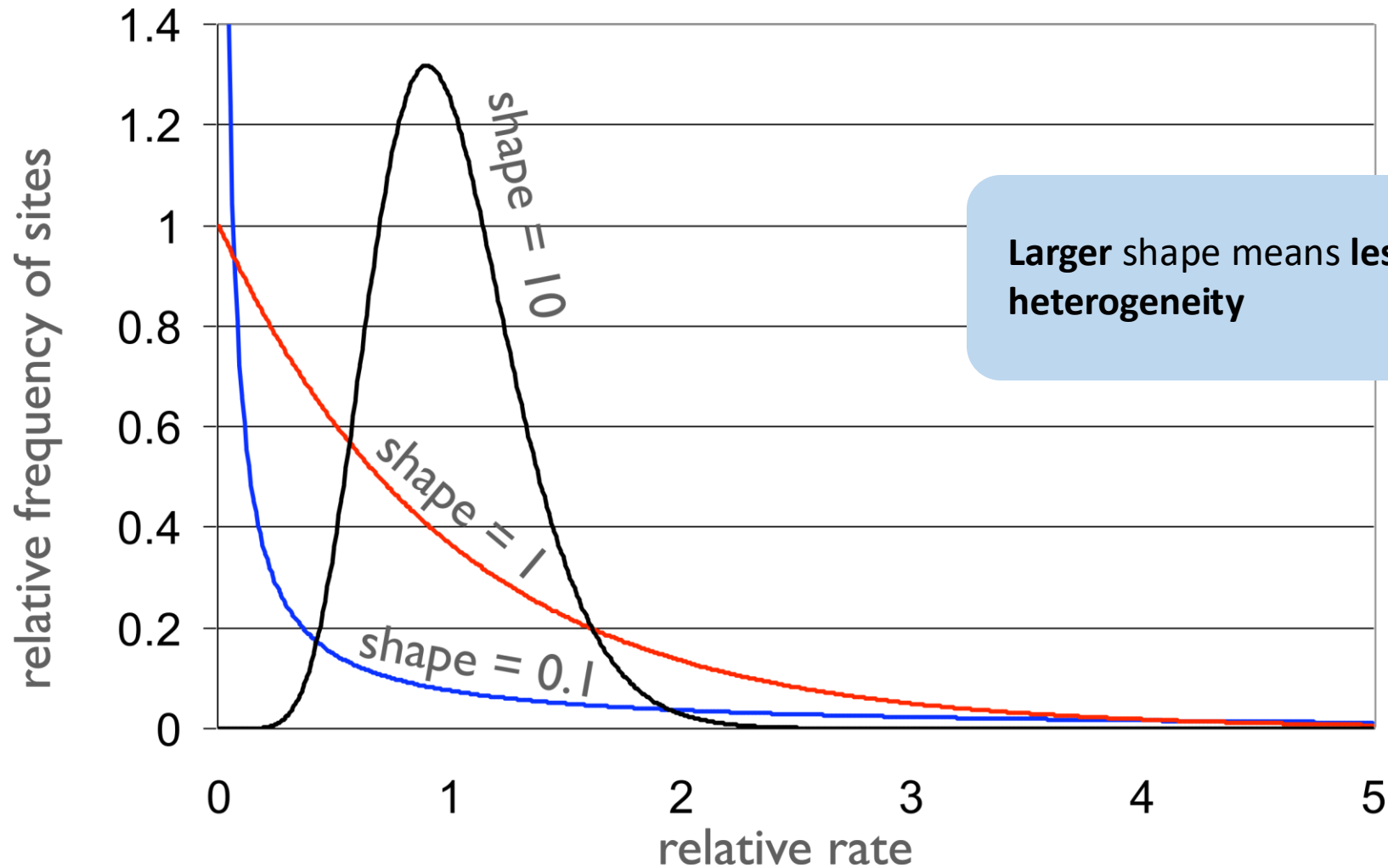
What do we do?

	T1	T2
Taxa A	0	0
Taxa B	0	1
Taxa C	1	2

Allow these traits to evolve at different rates:

- Specify which traits evolve fast
- **Use a gamma model to account for rate heterogeneity**

Across Site Rate Variation (+G)



Partitioning the data

Grouping together parts of the alignment that have similar characteristics and or may have **evolved together** due to evolutionary pressures

The **defaults** in many phylogenetic software is to group by maximum observed state size

$$Q = \begin{pmatrix} -\mu_0 & \mu_{01} & \mu_{02} & \mu_{03} \\ \mu_{10} & -\mu_1 & \mu_{12} & \mu_{13} \\ \mu_{20} & \mu_{21} & -\mu_2 & \mu_{23} \\ \mu_{30} & \mu_{31} & \mu_{32} & -\mu_3 \end{pmatrix},$$

$$\begin{bmatrix} -\mu_0 & \mu_{01} \\ \mu_{10} & -\mu_1 \end{bmatrix} \quad \begin{bmatrix} -\mu_0 & \mu_{01} & \mu_{02} \\ \mu_{10} & -\mu_1 & \mu_{12} \\ \mu_{20} & \mu_{21} & -\mu_2 \end{bmatrix}$$



Partitioning the data

When should we partition our data?

If we have presence (1) absence (0) traits partitioning will always be a logical approach: what would transitioning to state 2 in this scenario even mean?

We should be cautious for traits describing a trait – just because we do not observe a state 2 can we be absolutely certain there never was one?

Justifying partitioning schemes is very important as they have a major impact on inference results



Exercise 1:

Simulate morphological data using the Shiny app for MorphoSim

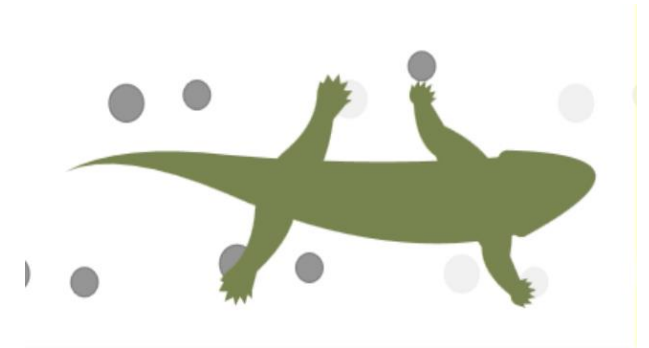
<https://github.com/fossilsim/MorphoSimShiny>

How do we choose a model and does it matter?

Used 114 empirical **tetrapod** matrices and compared 7 different models

Looked at the impact on:

- branch lengths (**evolutionary distances**)
- Tree topology (**species relationships**)



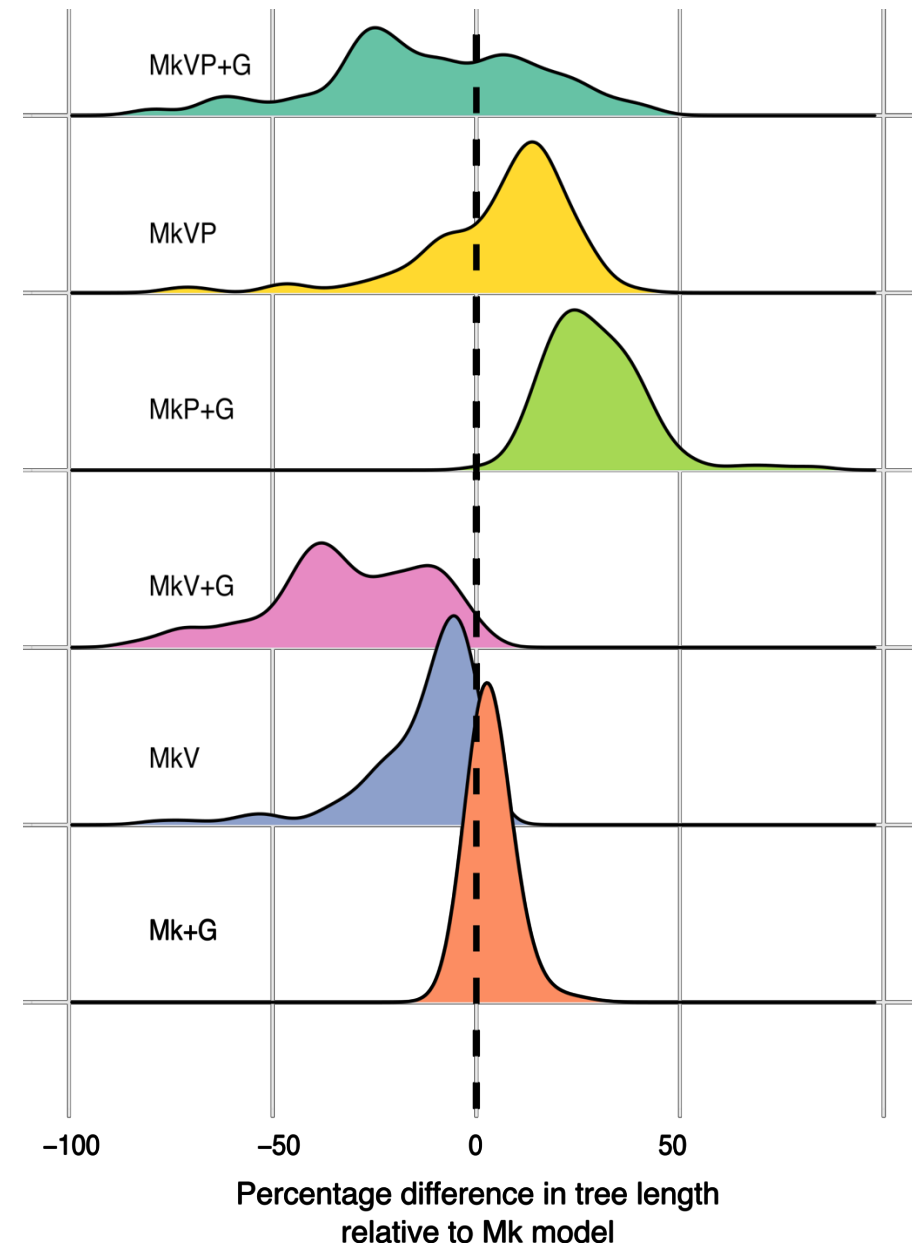
Impact of model choice

Used 114 empirical **tetrapod** matrices and compared 7 different models

Looked at the impact on:

- branch lengths (**evolutionary distances**)
- Tree topology (**species relationships**)

So how do we choose a model?





Model adequacy

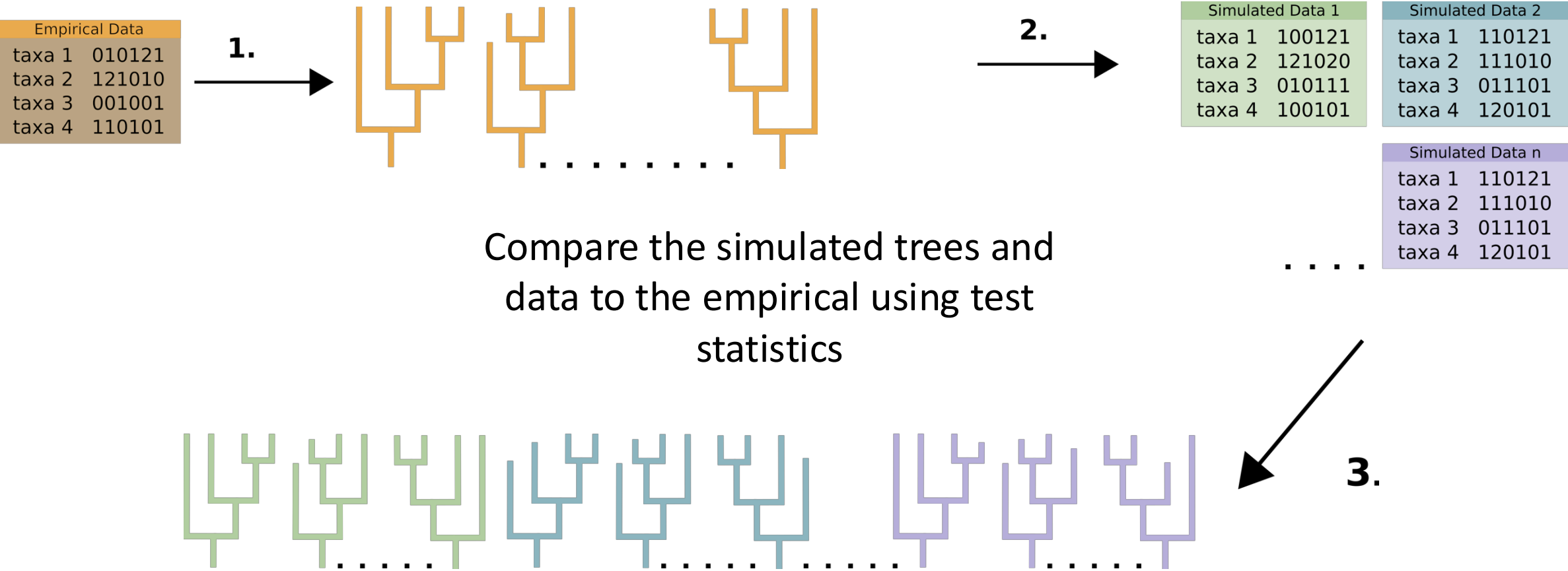
We know that none of our models are really true. Can we be sure that the chosen model captures the salient features of the evolutionary process and provides reliable inferences

Could the model and priors plausibly have given rise to the data

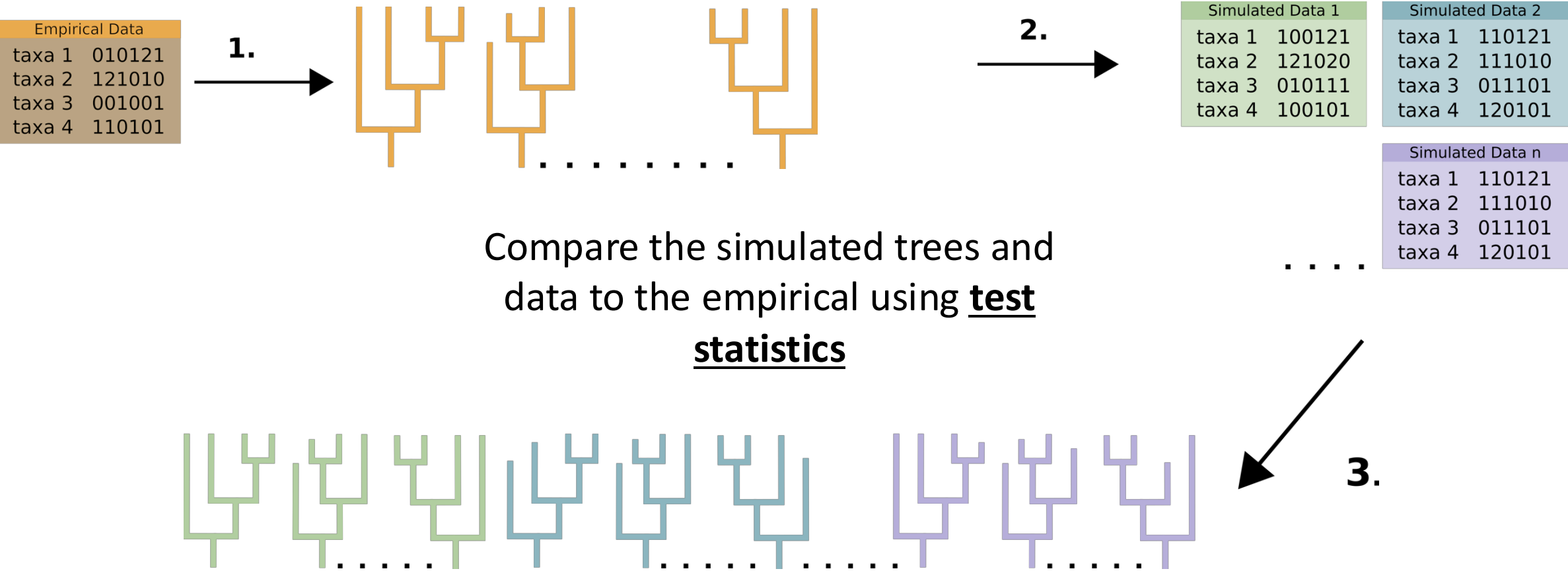
Allows us to ask whether **any** of our models are doing a good job describing the evolutionary processes that produced our data

Provides the **absolute fit** of our model to a data set

Posterior Predictive Simulations



Posterior Predictive Simulations





Test Statistics

A test statistic is a **numerical summary** of data.

A value that captures the characteristic of your data.

For PPS we use two test statistics: **Consistency Index** and **Retention Index**

These test statistics use both the data and the trees

Note: there may be more test statistics worth investigating but as of now these are the only two that are validated for use with morphological data, - see [Mulvey et al 2024](#) for more info



Exercise 2:

Carry out [posterior predictive simulations](#) for the data set in exercise one. Do either of the models fit our data?